

Reharvester: A Local-First System for Interactive Research-Literature Discovery

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Abstract. Keeping pace with a fast-growing research literature is increasingly difficult, and the cloud services that promise to help are opaque, non-portable, and require that sensitive reading lists leave the user's machine. We present Reharvester, a local-first system that turns a keyword or abstract query into an interactive map of a research field with no analysis delegated to an external service. Reharvester harvests bibliographic records, assembles them into a navigable knowledge graph, surfaces emerging keyword trends against an explicit null model, and exposes the corpus through a wiki-linked reader backed by retrieval-augmented generation over the graph. Bibliographic harvesting is the only networked step; indexing, graph construction, analysis, and question answering all execute on the user's hardware, and the system is organised as a capability ladder that degrades rather than fails when compute or models are unavailable. We evaluate a complete implementation on a 34,397-paper corpus spanning 2013–2026. The whole pipeline builds in 25s on a 16-core laptop, and the capability ladder retains 77% of full known-item recall using nothing but an inverted index. Hybrid retrieval with graph expansion reaches 0.958 recall@10 on known-item search and improves nDCG@10 by 0.047 (95% CI [0.032, 0.061]) over TF-IDF on a pooled-judgment topical task. Two results temper the design: graph expansion yields no significant gain in context quality over flat hybrid retrieval at equal token budget, and the exact k -nearest-neighbour backbone scales as $n^{2.01}$, which we address with an inverted-index pruning variant that scales as $n^{1.16}$ while retaining 82% of exact edges. The result is literature exploration that is reproducible, private, and air-gappable.

Keywords: Literature discovery · Knowledge graph · Local-first software · Retrieval-augmented generation · Research-gap analysis · Keyword trends.

1 Introduction

The volume of published research outpaces any individual's capacity to read it. A researcher entering an unfamiliar subfield faces two problems: finding the

handful of papers that matter, and building a mental model of how the subfield is organised—which communities exist, which are converging, which vocabulary has just appeared. Contemporary discovery tools address the first well and the second only incidentally.

They also centralise both the data and the reasoning. A query typed into a hosted service leaves the user’s machine, and so does the reading list it accumulates; for work on unpublished results or sensitive data, that egress is a reason not to use the tool at all. Centralisation also hides the ranking logic, so a result set cannot be audited, reproduced, or cited as a method: when a service changes its model, last year’s literature review is no longer reproducible, and nothing in the artefact records that it changed.

Local-first software [15] offers the alternative stance: the user’s device holds the primary copy of the data and remains functional without a network. We apply that stance to literature discovery. Reharvester reframes the task as a local, inspectable pipeline: after an initial harvest, graph construction, trend analysis, retrieval, and question answering all execute on the user’s hardware. The design commitment we make, and test, is that the system should never present a hard capability cliff: a user with a 90 MB neural encoder should get better results than a user without one, but the user without one should still get a working system rather than an error.

We contribute (i) a local-first architecture organised as an explicit *capability ladder*, in which each tier adds a dependency and every tier below remains a complete system; (ii) a trend-and-gap analytic layer pairing smoothed prevalence growth and burst detection with a *specificity* filter, scoring candidate gaps against a permutation null rather than a raw density threshold; (iii) an evaluation on 34,397 papers covering retrieval on two tasks with independent ground truth, context quality at fixed budget, degradation, and scaling with measured exponents; and (iv) two negative results that constrain the design—graph expansion does not improve retrieval-augmented context at equal budget, and the exact backbone is quadratic, for which we give a near-linear replacement.

2 Related Work

Science mapping. VOSviewer [6] and CiteSpace [4] established the interactive co-citation map as the standard artefact. These are desktop applications, local in the trivial sense, but they consume curated citation databases and treat the map as the endpoint. Reharvester treats it as an index into a reader: the graph is the substrate for retrieval and question answering, not only for visual inspection. Open corpora make this possible—S2ORC [18] and OpenAlex [20]—though our ingestion stage is thin, and any source yielding titles, abstracts, authors, and dates can drive the pipeline.

Retrieval. Sparse lexical retrieval remains a strong baseline, with BM25 [22] the reference method. Dense retrieval [13,10] now dominates; scientific-domain [1] and general sentence encoders [21] supply the embeddings, benchmarks [23] govern model selection, and approximate nearest-neighbour indexes [19] make dense search tractable. Hybrid systems combine both signals, most simply through

reciprocal rank fusion [5]. Our contribution is not a new retriever but a measurement of how these components behave when the constraint is one consumer machine and every tier must still work without the one above it.

Retrieval-augmented generation. RAG [16] grounds generation in retrieved evidence, mitigating the long-tail knowledge gaps of parametric models [12]; refinements address long-context position sensitivity [17] among much else, open-weights models [11] make local generation feasible, and automated evaluation [8] informs our metric design. Most relevant is graph-structured RAG: GraphRAG [7] builds an entity graph for query-focused summarisation and HippoRAG [9] uses a graph as associative memory; both report that graph structure helps. We test the same idea for a document-level semantic backbone under a fixed token budget and—contrary to our expectation—find no significant benefit (Section 6).

Trends, gaps, and reproducibility. Burst detection in document streams [14] locates the moment a topic emerges. Structural-hole theory [3] motivates reading sparse regions between dense communities as opportunity, and community detection [2] provides the machinery. Our analytic layer adds two guards absent from the tools we surveyed: a specificity filter and an explicit null model. Separately, both FAIR data practice and systematic-review reporting standards require a search to be describable and repeatable, which a hosted service whose model changes silently cannot satisfy.

3 System Design

Reharvester is a five-stage pipeline (Fig. 1a) in which each stage writes a durable local artefact, so any stage can be re-run, inspected, or replaced without re-running those before it.

Harvest and index. Ingestion pulls title, abstract, authors, categories, and date from an open API into a flat local store, deduplicating on the version-stripped identifier; this is the only stage that touches the network, and afterwards the system is fully functional offline. It retrieved 37,006 records in 928 s, a rate set by the source API’s politeness delay rather than local compute. Three indexes are then built over abstracts: a TF-IDF index (unigrams and bigrams, sublinear term-frequency scaling) that supports cosine ranking and supplies the vector space for the graph; a BM25 index [22] with different failure modes; and a dense index encoding each abstract into 384 dimensions with a compact sentence encoder [21]. These cost 5.0 s, 1.6 s, and 388 s respectively; the dense index is by far the most expensive stage, which is why the ladder treats it as optional.

Graph. The semantic backbone is a mutual k -nearest-neighbour graph over the TF-IDF space with $k = 8$, edges weighted by cosine similarity, yielding 203,100 edges. A co-authorship layer adds 269,295 edges over 71,988 distinct author strings, with authors on more than 60 papers excluded to limit name-collision effects. Louvain community detection [2] partitions the backbone into 35 communities, each labelled by its centroid’s highest-weight terms, and a per-node bridge score—the fraction of a node’s edge weight crossing a community boundary—identifies papers between topics, operationalising the structural-hole

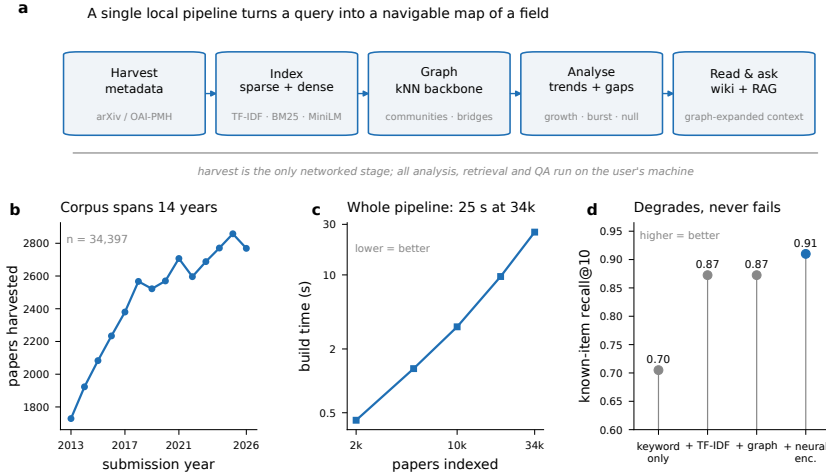


Fig. 1: System overview. (a) The five pipeline stages; harvesting is the only stage that contacts the network, and everything downstream of it runs locally. (b) The evaluation corpus, harvested from five arXiv categories, spans 2013–2026 ($n = 34,397$ after deduplication and abstract-length filtering). (c) End-to-end build time for indexing, graph construction, community detection and trend analysis on a 16-core Apple-silicon laptop. (d) Known-item recall@10 across the capability ladder ($n = 400$ queries): each tier adds one dependency, and the tier with no model and no graph still answers 70.5 % of queries.

intuition [3] at document level. The backbone, not the raw index, makes the system navigable: it converts a ranked list into a neighbourhood the user can walk.

Analyse. This layer answers two questions a ranked list cannot: what vocabulary is emerging, and where are the gaps? Both required care, and both went through a failed first version we report because the failure is instructive.

A naive n -gram extractor that removes stopwords and then forms bigrams from the surviving token stream generates phrases that never occurred in the text: in our first run “models llms” was the fastest-rising bigram in the corpus with 1,579 late-window occurrences, an artefact of joining “... models.” to “LLMs have...” across a sentence boundary. Our extractor therefore splits on sentence and clause punctuation first, forms n -grams only within maximal runs of non-stopword tokens, and discards residual L^AT_EX control words, which otherwise appear as burst terms (`textbf` peaked in 2026 with burst weight 16.4).

Prevalence growth compares a term’s document-frequency share between an early and a late window. An unsmoothed log-ratio is infinite for any term absent from the early window—exactly the interesting case—so we smooth with $\alpha = 0.5$ pseudo-documents, taking $\text{lift}(t) = [(n_{\text{late}}(t) + \alpha)/D_{\text{late}}] / [(n_{\text{early}}(t) + \alpha)/D_{\text{early}}]$ where $n_w(t)$ counts documents in window w containing t and D_w counts documents in that window. Raw counts are reported alongside so a reader can distinguish a new phrase from a merely growing one.

Burst and growth statistics are otherwise dominated by discourse vocabulary (“underexplored”, “findings highlight”) that rises because writing conventions change, not because a topic emerged. We filter on the normalised entropy of a term’s distribution over communities, $S(t) = -\sum_c p(c | t) \log p(c | t) / \log |C|$, retaining terms with $S \leq 0.80$: a term spread evenly over all communities has $S \approx 1$, a topical term concentrates. The median unigram specificity here is 0.778, so this is a mild filter that nonetheless removes the worst offenders.

Finally, a research gap is a pair of communities whose documents read alike but between which few links exist. Scoring this as raw cross-edge density conflates a real gap with the fact that small communities have few edges: our first version ranked a small recipe-and-food community against nearly everything else. We instead compare the observed cross-community edge count against a null obtained by permuting community labels over the fixed graph 200 times—preserving the degree sequence and the community sizes—and report a z -score, taking as candidates the pairs with high centroid cosine similarity and $z < -2$.

Read and ask. The reader presents papers with wiki-style links along backbone edges, so navigation follows semantic adjacency rather than a ranked list. Question answering assembles a context set under a fixed token budget—seeding with hybrid retrieval, expanding along the backbone—and hands it to a local open-weights model [11]. We evaluate the context assembly rather than the generated prose, since context quality is the part our design choices control.

3.1 The capability ladder

A local-first system runs on hardware it does not choose. Reharvester is therefore built as four tiers, each a complete system: **T0**, an inverted index only, with boolean candidate generation and token-overlap ranking, needing nothing beyond the standard library; **T1**, adding a TF-IDF vectoriser; **T2**, adding the k -NN backbone, which enables neighbourhood expansion and the community and trend analytics; and **T3**, adding a neural sentence encoder (≈ 90 MB). The ladder is not a fallback path taken on error but a configuration the user chooses.

4 Implementation and Evaluation

The implementation is Python 3.13 with NumPy, SciPy, scikit-learn, NetworkX, and `rank_bm25`; the optional dense tier uses `sentence-transformers` with a 384-dimensional MiniLM encoder. No component requires a GPU, and none issues a network request after harvesting. All measurements were taken single-process on a 16-core Apple-silicon laptop (macOS 26.5, 137 GB RAM). Two choices deserve note: retrieval indexes are built over *abstracts only*, which makes the known-item task below a genuine test at some cost in absolute accuracy; and the graph is built from the TF-IDF space rather than the dense space, so the backbone, the communities, and every analytic result depending on them remain available at tier T2 with no neural model present.

Corpus. We harvested five arXiv categories—cs.IR, cs.DL, cs.CL, cs.SI, cs.DB—across 2013–2026, retaining records with abstracts longer than 200 characters; after deduplication this gives 34,397 papers (Fig. 1b), 1,729–2,858 per year. The

corpus is deliberately heterogeneous, spanning lexical retrieval, network science, and database systems, so community structure is present but not trivially separable.

Ground truth. Evaluating a discovery system without relevance assessors requires ground truth the system under test did not produce; we use three sources, none derived from the machinery being measured. In **Task A** (known-item retrieval) the query is a paper’s title and the single relevant document is its own abstract; because titles are excluded from every index, this measures whether a retriever can bridge a short, differently-worded query to the right document. In **Task B** (topical retrieval) the query is a paper’s abstract and relevance follows arXiv’s own multi-label category assignment, applied by authors and moderators independently of our system: grade 3 for the same primary category and ≥ 2 shared labels, 2 for the same primary category, 1 for ≥ 2 shared secondary labels, 0 otherwise. Sharing a single broad label is not relevant, which keeps the task discriminative, and judgments are *pooled*—the ideal nDCG ranking comes from the union of all five systems’ top-50 results, so no system is scored against a reference derived from its own output. In **Task C** (context quality) a gold set of five documents per seed is defined by proximity in a *character-4-gram* TF-IDF space—used by no retriever and not involved in building the graph—further constrained to grade ≥ 2 under the Task B criterion. This metric is the one most at risk of circularity, and the independent judge space is what removes it: our first version defined gold sets from graph neighbourhoods and consequently reported that graph expansion won. Confidence intervals are paired bootstrap with 5,000 resamples.

5 Results

5.1 Retrieval quality

Table 1: Retrieval quality. Task A is known-item search over 500 queries; Task B is topical search over 400 queries with pooled graded judgments. $P@10_{\geq 2}$ counts only high-relevance hits (grade ≥ 2 of 3). Latency (ms/q) is mean wall-clock time per query, single-threaded, including all constituent retrievals for hybrid methods. Best value per column in bold.

Method	Task A: known-item			Task B: topical				ms/q
	P@1	R@10	MRR	nDCG@10	P@10	P@10 $_{\geq 2}$	R@20	
TF-IDF	0.612	0.866	0.701	0.443	0.599	0.484	0.581	5.3
BM25	0.758	0.918	0.817	0.435	0.588	0.475	0.568	32.2
Dense (MiniLM)	0.700	0.924	0.779	0.485	0.669	0.532	0.659	2.1
Hybrid (RRF)	0.760	0.950	0.837	0.476	0.642	0.521	0.610	39.7
Hybrid + graph	0.762	0.958	0.839	0.490	0.666	0.536	0.645	39.7

On known-item search (Table 1, Fig. 2a) hybrid retrieval with graph expansion reaches 0.958 recall@10, ahead of rank fusion alone (0.950), dense retrieval (0.924), and BM25 (0.918); the ordering is unsurprising and the graph contribution small. Topical search is more informative. Hybrid-plus-graph attains the

best nDCG@10 (0.490), a paired-bootstrap difference against the TF-IDF baseline of +0.047 with 95 % CI [0.032, 0.061]. Dense retrieval alone attains 0.485 (+0.041, CI [0.023, 0.060])—statistically indistinguishable from the full hybrid pipeline—at 2.1 ms per query against 410.0 ms, a factor of about 200. BM25 is the only method that does not significantly beat TF-IDF here (−0.009, CI [−0.022, +0.005]) despite being clearly better on known-item search: a lexical ranker tuned for short queries does not transfer to a task whose query is a 200-word abstract. The practical reading is that on any machine able to run the neural encoder, dense retrieval alone is the right default, and the hybrid pipeline earns its 200-fold latency cost only where the last two points of nDCG matter.

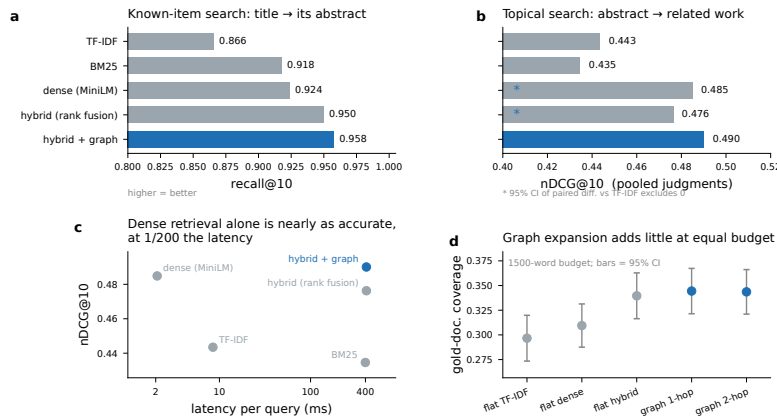


Fig. 2: Retrieval and context results. (a) Known-item recall@10, 500 queries. (b) Topical nDCG@10, pooled graded judgments, 400 queries; asterisks mark methods whose paired-bootstrap 95 % CI against TF-IDF excludes zero. (c) The quality–latency trade-off. (d) Gold-document coverage under a 1,500-word budget, 250 seeds, 95 % CIs.

5.2 Context quality: a negative result

Task C tests the assumption behind graph-expanded RAG. Under a 1,500-word budget, flat hybrid retrieval captures 0.340 of the gold set, one-hop graph expansion 0.344, and two-hop 0.344; the paired-bootstrap difference for one-hop expansion over flat hybrid is +0.005, CI [−0.006, +0.015], no significant effect. Both flat single-retriever baselines are significantly *worse* than flat hybrid (TF-IDF −0.043, CI [−0.061, −0.025]; dense −0.030, CI [−0.049, −0.012]), so the experiment has enough power to detect differences of the size we hoped to see from the graph. A secondary measurement shows the mechanism: averaged over seeds, flat TF-IDF context spans 4.00 distinct communities, flat hybrid 3.07, and one-hop expansion 2.73. Expansion along a semantic backbone retrieves documents similar to those already retrieved, so at fixed budget it trades topical diversity for local density—a trade that is approximately neutral when the gold set is defined independently of the graph.

Table 2: (a) Capability ladder on known-item retrieval, 400 queries; each tier adds one dependency to the one above it. (b) End-to-end scaling, single process, with peak RSS measured in an isolated subprocess per corpus size and exponents from a log-log fit across the five sizes.

(a)				(b)					
Tier Adds		R@10	P@1	Papers	Index	k -NN	Louv.	Total	RSS
					(s)	(s)	(s)	(s)	(MB)
T0	inverted index	0.705	0.600	2,000	0.22	0.05	0.08	0.42	591
T1	+ TF-IDF	0.873	0.635	5,000	0.58	0.32	0.22	1.30	1,075
T2	+ k -NN graph	0.873	0.635	10,000	1.24	1.13	0.49	3.24	2,562
T3	+ encoder	0.910	0.738	20,000	2.73	4.85	1.31	9.65	3,401
				34,397	5.61	15.79	2.54	25.34	3,626
Exp.					$n^{1.13}$	$n^{2.00}$	$n^{1.21}$	$n^{1.43}$	$n^{0.69}$

5.3 The capability ladder

The tier with no vector space, no graph, and no model answers 70.5 % of known-item queries within the top ten—77 % of what the full neural tier achieves (Table 2a). Adding the TF-IDF vectoriser recovers most of the remaining gap (0.873), and the neural encoder adds a further 0.037 recall and a substantial 0.103 in P@1. Notably T3 is also the *fastest* tier at query time (2.1 ms), because a 384-dimensional dense scan is cheaper than a sparse cosine over a 120,000-term vocabulary; its cost is paid once, at index time, as the 388 s encoding pass. Tier T2 does not change known-item accuracy at all, which is the honest reading: the backbone exists to support navigation, communities, and the trend analytics, and should not be justified by retrieval numbers.

5.4 Trends and gaps

The fastest-rising bigrams over 2022–26 relative to 2013–17 (Fig. 3a) are “graph neural” (smoothed $\log_2$ lift 9.26, 404 late-window documents), “large language” (9.00, 2,374), “retrieval-augmented generation” (8.91, 317), and “contrastive learning” (8.36, 216); the steepest declines are “distributed representations” (−4.74, 50 documents to 2), “statistical machine” (−4.37, 70 to 4), and “thomson reuters” (−6.38, 31 to 0)—the last a vendor name whose disappearance tracks a database rebranding rather than a scientific shift, a reminder that the method detects vocabulary, not ideas. Trajectories make the substitution legible: “statistical machine” peaks in 2015 at 0.96 % of abstracts and reaches 0 % by 2024, while “graph neural” peaks at 3.87 % in 2023–24 and “retrieval-augmented generation” rises from zero to 5.05 % in 2025. The specificity filter makes this readable: without it the top burst terms are “prompt” (legitimate) followed by “decouples”, “aligns”, and “underexplored”.

Fig. 3c–d show the coarse-grained map—25 communities of at least 400 papers, 133 inter-community links above threshold—and the top gap candidates. The highest-similarity under-connected pairs are the two network-science communities (centroid cosine 0.736; 927 cross-links against a null mean of 1,015, $z = -2.6$), question answering against large language models (0.696; 697 vs.

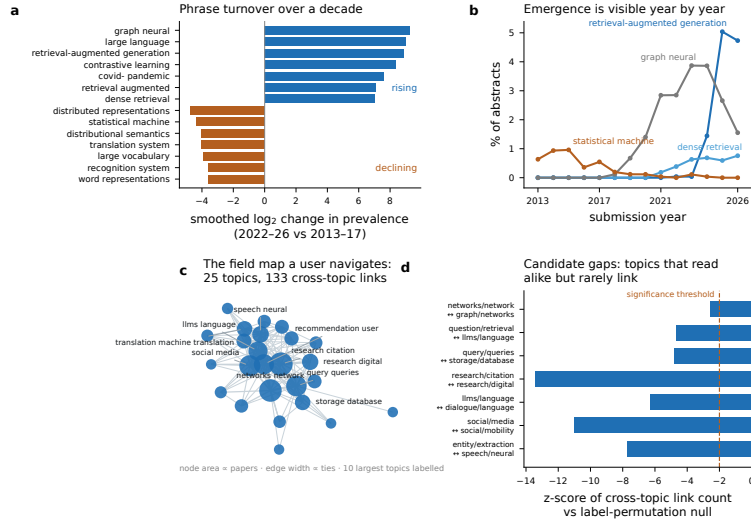


Fig. 3: The analytic layer. (a) Seven fastest-rising and seven steepest-declining bigrams by smoothed $\log_2$ prevalence change (2022–26 vs 2013–17), after specificity filtering. (b) Year-by-year prevalence of four phrases, showing substitution rather than monotone growth. (c) The coarse-grained field map: 25 communities of ≥ 400 papers, edges for inter-community links above threshold, ten largest topics labelled. (d) The seven highest-similarity pairs whose cross-link count falls below a label-permutation null; because 536 of 561 pairs cross $z < -2$, similarity rather than significance drives this ranking.

833, $z = -4.7$), query processing against storage systems (0.688; 892 vs. 1,064, $z = -4.8$), and citation analysis against digital-library infrastructure (0.672; 1,045 vs. 1,616, $z = -13.5$). Each names two communities that share vocabulary and methods but publish in different venues. One limitation must be explicit: of 561 pairs, 536 reach $z < -2$, because permuting labels destroys all topical structure and so makes nearly every real pair look under-connected. The z -score is thus a weak filter and the ranking is carried almost entirely by centroid similarity; a stronger null would preserve topical assortativity, and we regard this result as a shortlist generator rather than a hypothesis test.

5.5 Scalability

The whole pipeline builds in 25.3s at 34,397 papers within 3.6 GB of resident memory (Table 2b), with indexing, community detection, and trend analysis all near-linear ($n^{1.13}$, $n^{1.21}$, $n^{1.05}$). The exact k -NN backbone is not: it scales as $n^{2.00}$, consistent with its brute-force all-pairs formulation, and dominates total cost beyond about 10,000 papers—extrapolating the fit, a 500,000-paper corpus would need roughly 53 min for the backbone alone.

We therefore implemented an approximate backbone that prunes candidates using the TF-IDF inverted index: for each document only its 24 highest-weight terms are probed, rarest postings first, and the candidate pool is capped at 400

before exact cosine scoring. This needs no external ANN library, preserving the local-first property. The variant scales as $n^{1.16}$ against $n^{2.01}$ for the exact computation, crossing over at roughly 20,000 papers and reaching a $1.97\times$ speedup at 34,397 (8.25 s vs. 16.24 s). The fidelity cost is real and grows with corpus size: edge recall against the exact graph falls from 0.941 at 5,000 papers to 0.815 at 34,397, and the adjusted mutual information between the resulting community partitions falls from 0.749 to 0.624. The approximation is therefore appropriate for navigation and retrieval seeding, and questionable for community-level analytics, where we default to the exact backbone.

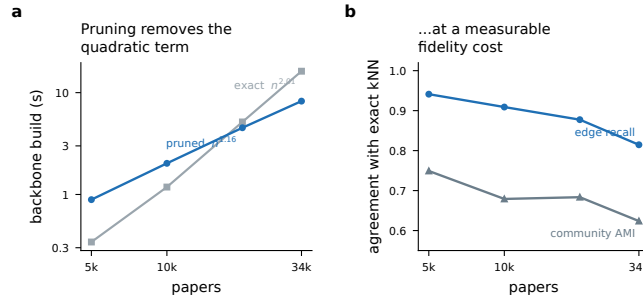


Fig. 4: The backbone bottleneck. (a) Inverted-index candidate pruning reduces the scaling exponent from $n^{2.01}$ to $n^{1.16}$, crossing over near 20k papers. (b) The cost: edge recall and community agreement (AMI) against the exact backbone both degrade as the corpus grows. Per-stage wall times are in Table 2b.

6 Discussion

Three claims survive our own attempts to break them. A full literature-mapping pipeline over tens of thousands of papers is a 25-second local computation, not a service. The capability ladder is real: the bottom rung retains 77% of top-tier known-item recall with no model and no graph, so a user on constrained hardware or inside an air gap gets a working system. And on any machine that can run a 90 MB encoder, dense retrieval alone is the right default, within 0.005 nDCG of the full hybrid pipeline at 1/200 of the latency.

Our negative result on context quality differs from reports that graph structure improves RAG [7,9], and the difference is instructive rather than contradictory. Those systems build graphs over *entities* and use them to assemble evidence that is topically distant but relationally connected; our backbone connects *whole documents* by lexical similarity, so expanding along it moves toward documents resembling those already retrieved, which at fixed budget reduces topical diversity (3.07 to 2.73 communities per context) without improving coverage. The lesson is not that graph-structured RAG fails but that the graph must encode a relation the retriever does not already capture: a citation or entity layer satisfies that condition, a semantic k -NN layer over the retriever’s own features does not.

Limitations. Ground truth comes from arXiv category labels and an independent lexical space, not human assessors; both are proxies, and category grading

in particular rewards retrieving papers from the same subfield rather than useful ones. Our corpus is five computer-science categories from one repository, so community structure may be more separable than in an interdisciplinary corpus, and the gap null model is weak as Section 5.4 states. We evaluate context assembly rather than generated answers, so no claim is made about end-to-end answer quality; doing so would need human judgment or an LLM-based protocol [8] whose own validity would need establishing. All timings come from one machine class.

Future work. The most valuable addition is a citation layer, which our own negative result predicts would make graph expansion effective; beyond that, a stronger null preserving topical assortativity, incremental index and graph updates so a weekly harvest does not force a rebuild, and a human study, since every claim here about what helps a researcher navigate a field is currently an inference from automatic metrics.

7 Conclusion

Reharvester shows that interactive research-literature discovery does not require a cloud service. A five-stage pipeline over 34,397 papers builds in 25 s on a laptop, produces a navigable field map with 35 communities and an auditable trend analysis, and answers known-item queries at 0.958 recall@10. Organising the system as a capability ladder rather than a monolith means it degrades instead of failing: 70.5 % recall@10 survives with nothing but an inverted index. Two measured negatives shape the design as much as the positives—graph expansion over a document-similarity backbone does not improve retrieval-augmented context at equal budget, and the exact backbone is quadratic, needing the near-linear pruning variant we give. Because everything after harvesting runs locally and every artefact is inspectable, a literature search performed with Reharvester is something a researcher can hold, repeat, and cite.

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